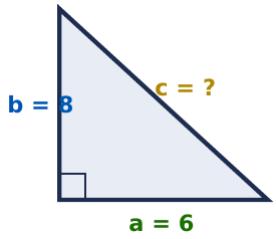
Unit 5 Classwork — Pythagorean Theorem — Answer Key

Hypotenuse · Missing Leg · Converse · Word Problems · Coordinate Distance

A. Find the Hypotenuse

Problem 1 Score 3

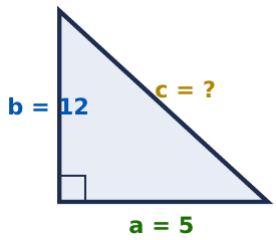
Find the hypotenuse (c) of the right triangle shown.



Answer: 10

Problem 2 Score 3

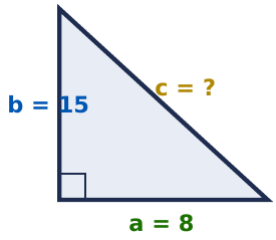
Find the hypotenuse (c) of the right triangle shown.



Answer: 13

Problem 3 Score 4

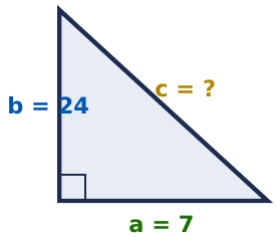
Find the hypotenuse (c) of the right triangle shown.



Answer: 17

Problem 4 Score 4

Find the hypotenuse (c) of the right triangle shown.

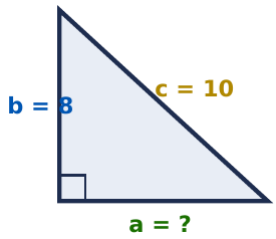


Answer: 25

B. Find a Missing Leg

Problem 1 Score 3

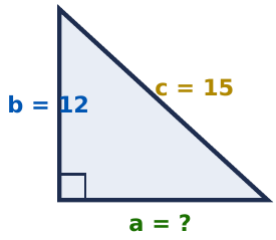
Find the missing leg (a) of the right triangle shown.



Answer: 6

Problem 2 Score 4

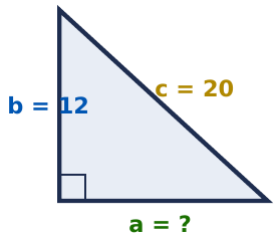
Find the missing leg (a) of the right triangle shown.



Answer: 9

Problem 3 Score 4

Find the missing leg (a) of the right triangle shown.

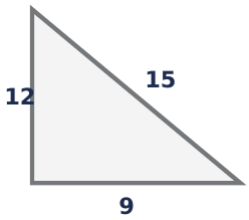


Answer: 16

C. Converse — Is It a Right Triangle?

Problem 1 Score 3

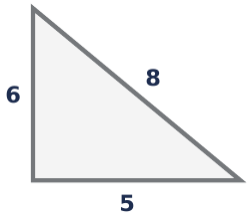
The triangle shown has sides 9, 12, 15. Is it a right triangle?



Answer: yes

Problem 2 Score 4

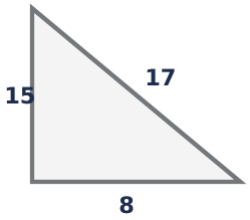
The triangle shown has sides 5, 6, 8. Is it a right triangle?



Answer: no

Problem 3 Score 4

The triangle shown has sides 8, 15, 17. Is it a right triangle?

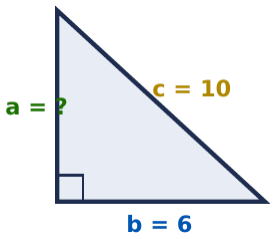


Answer: yes

D. Word Problems

Problem 1 Score 4

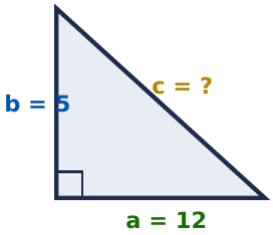
A 10-foot ladder leans against a wall. The base is 6 feet from the wall. How high up the wall does the ladder reach?



Answer: 8

Problem 2 Score 5

A rectangular garden is 12 m long and 5 m wide. What is the length of the diagonal?

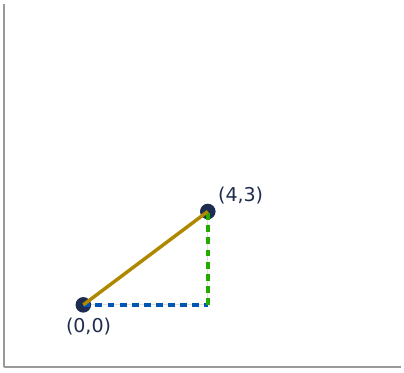


Answer: 13

E. Distance on the Coordinate Plane

Problem 1 Score 4

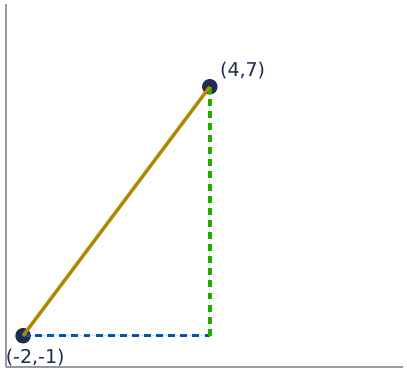
Find the distance between the two points shown.



Answer: 5

Problem 2 Score 5

Find the distance between the two points shown.

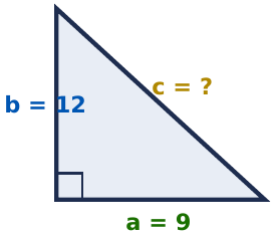


Answer: 10

F. Mixed Review

Problem 1 Score 4

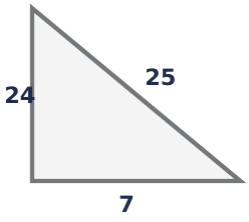
Find the hypotenuse (c) of the right triangle shown.



Answer: 15

Problem 2 Score 4

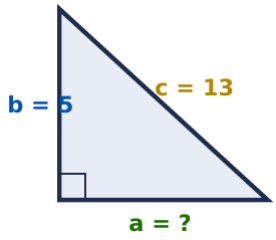
The triangle shown has sides 7, 24, 25. Is it a right triangle?



Answer: yes

Problem 3 Score 5

Find the missing leg (a) of the right triangle shown.



Answer: 12